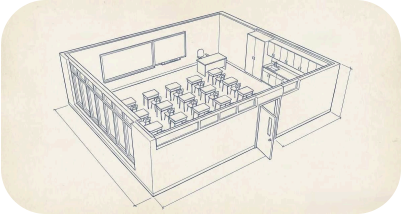
Analysis of the effect of class size on student academic performance



2026 STA207 Final Project

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👉 **PRESS TO START**

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👉 Next: Background

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Background

The Student Teacher Achievement Ratio study was a randomized experiment conducted over the course of 4 years (1985 - 1989) at various schools in Tennessee. The main goal of the experiment was to investigate the effect of class size on student performance in Kindergarten through 3rd grade. Beginning in Kindergarten, participating students were randomly assigned into one of three treatment groups based on class size: small class (13 - 17 students), regular class (22-25 students) and regular class plus aide which is the same size as the regular class group but with an additional teacher's aide. Within each school, teachers were also randomly assigned across class size groups. Academic output was tracked using a number of metrics including standardized (Stanford Achievement Test, referred to as SAT) and curriculum-based (Basic Skills First, referred to as BSF) test scores in various subjects, mastery of objectives, and a motivation score. A large amount of covariate data was also recorded based on students, schools, and teachers which gives us access to additional useful information that we can potentially implement as precision variables in our analysis. It is important to note that the precision variables are not necessary to establish causality as this is already handled by random assignment. However, if we account for variables related to ethnic and socioeconomic background of students, teacher experience and credentials, and school location (among others), we can possibly improve the precision and accuracy of the treatment effect estimates by explaining more of the variation in academic performance.

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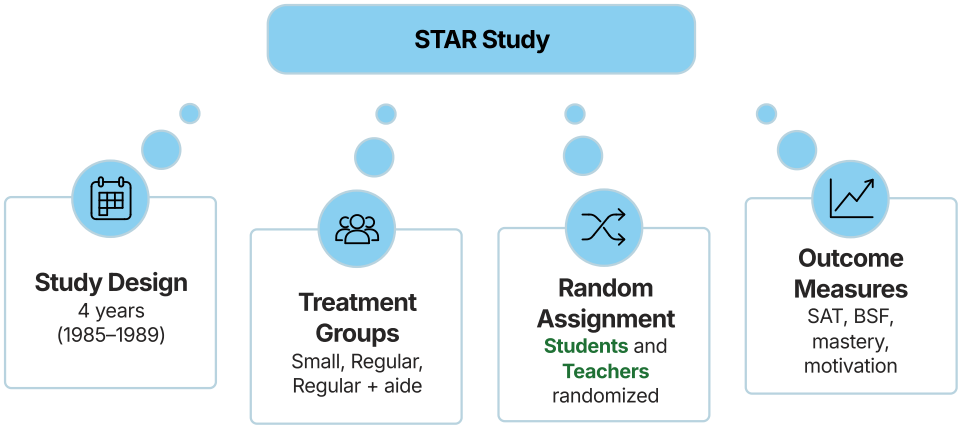

👉 **Next: Question of Interest**

Questions of Interest

Does the assigned Kindergarten class type have an effect on Kindergarten Math SAT scores?

Challenges to Causality

- Non-stayers (1-3 years

SUTVA Violations

- Consistency: Class size variation

Our Solution: Focus on Kindergarten

- Closest to

Why Math Scores?

- Classroom environment critical

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vs 4 years) • Non-compliers (students switching groups) • Confounding imbalance	(13-17, 22-25) • Interference: Spillover effects between students • Biased causal estimates	original random assignment • Minimal switching/reassignment • Strongest causal interpretation • One year of clean exposure	for math • Hands-on learning in class • Teacher assistance essential • Best indicator of class size effect

Question(s) of Interest

Does the assigned Kindergarten class type have an effect on Kindergarten Math SAT scores?

In this experiment, the random assignment of both students and teachers is key in balancing confounders among treatment groups and theoretically establishing causality between class size and academic metrics. We say theoretically because there are some notable issues that make it more complicated to e

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causality. First of all, random assignment is disrupted by both non-stayers and non-compliers. Non-stayers are defined as students who are only part of the experiment for one to three years, instead of the full four years. Non-compliers are students who switch treatment groups during the experiment. By overriding the original random assignment, these groups create an imbalance of the confounders and prevent a traditional causal analysis for the entire experiment. This is because compliance and staying is not random: there are systematic differences between compliers and non-compliers, stayers and non-stayers that can affect outcomes. For instance, it is possible that non-compliers were switched to a smaller class size at their parents' request, and having parents who care enough to do that is a factor that might be associated with higher test scores (maybe they are also hiring a tutor on the side.)

Adding onto this, SUTVA is violated when looking at the experiment over multiple years. The first component of SUTVA – consistency of treatment across all units in a treatment group – is violated due to the fact that treatment groups are based on ranges of class sizes (13-17, 22-25) so there is variation across small, regular, and regular with aide classes when it comes to number of students. Non-compliers/switchers and non-stayers also ensure that class sizes are fluctuating year to year, and this may even cause some classes to go from small to regular or vice versa. The second component of SUTVA – no interference between subjects – is also broken because there is interaction between students that occurs outside of the classroom as well as within the classroom between students who have switched classes (spillover effect.) Overall, these violations can result in biased causal estimates.

Based on Project STAR's limitations outlined above, we decided to only focus on kindergarten. Our reasoning behind this is that it is the stage of the experiment that is most closely aligned with the original random assignment whereas the outcomes for

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later grades are more vulnerable to switching, reassignment, class size fluctuations, and cumulative exposure differences. Thus, narrowing our scope to kindergarten preserves the cleanest treatment to outcome link. In turn, this provides us with the strongest basis for causal interpretation of the actual treatment on the entire sample as opposed to the effect of treatment assignment (which can be estimated via ITT). One thing to note is that the immediate effects make be modest as subjects only have one year of exposure to the treatment.

With regards to our response variable, we opted to go with math scores since intuitively we believe math is the subject where the classroom and teaching environment typically play the largest role. Whereas our concept of the English language can be learned both actively and passively, math is a topic that is most effectively learned in class via hands-on activities and assistance from teachers. Hence, since we assume a higher proportion of learning is done in class for math compared with other subjects, our logic is that focusing on standardized scores such as the SAT math scale gives us the best indication of whether class size had an effect on student performance.

👉 **Next: Descriptive Analysis**

Descriptive Analysis

Dataset Overview

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The dataset used in this analysis comes from the Project STAR experiment, which follows students from kindergarten through later grades. The dataset contains detailed information about students, teachers, and schools, including demographic characteristics, identifiers for instructional groups, and academic performance measures. The initial dataset comprises 11,601 observations across 379 variables.

For this exploratory analysis, we focus on kindergarten total SAT math scores, which reflect students' early academic preparation. Several variables are considered in the analysis, including school ID, teacher ID, race, and free-lunch status. We filtered out observations with missing values in gktmathss, gktchid, and gkfreelunch. After this data cleaning step, 5,853 observations remained, and the following analysis is based on this cleaned dataset.

Among these variables, school ID and teacher ID represent hierarchical grouping structures, while race and free-lunch status represent students' family background characteristics that may influence academic performance.

1. School and Kindergarten Math Scores

The boxplots indicate that the median math scores across schools are relatively similar, but the variability within schools is slightly different. Some schools show a wider interquartile range and more extreme observations.

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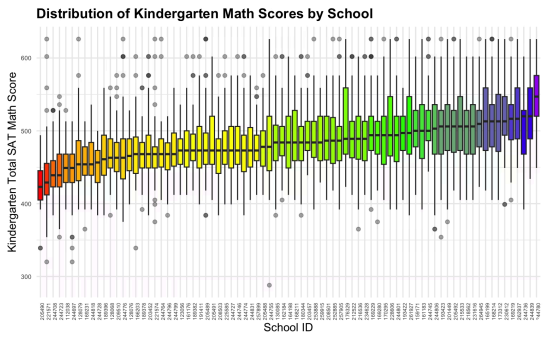

Although no strong pattern emerges, the differences in spread and the presence of outliers across schools indicate that school-level variation may exist. Since students are naturally clustered within schools, ignoring this structure may underestimate the variability in student performance.

Therefore, school ID represents a potential group-level source of variation that should be considered in the modeling stage. And we wanted to include school ID as a **fixed effect** in the model.

2. Teacher ID and Kindergarten Math Scores

Table: Summary Statistics of a Sample of Teacher IDs

gktchid	mean	median	min	max	Q1	Q3
11203801	438.7	429	399	500	415	458.5
11203802	440.7	439	392	484	424.5	462
11203803	469.3	463	429	528	446.5	494

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12305601	472.1	468	412	528	454	489
12305602	478.9	484	429	520	465.5	492
12305603	473.9	473	429	536	454	491.5
12806801	465.5	454	434	536	449	473
12806802	467.9	459	412	559	440.2	487.8
12806803	473.2	478	384	547	456.5	491.5
12807601	468	463	439	513	449	484

Teacher ID is the closest variable we have to class ID, because knowing a student's teacher usually tells us which class the student was in. Students in the same class may be correlated since they share the same teacher, classroom environment, and peer interactions.

On this page, we present summary statistics of kindergarten math scores by teacher ID. From the table, we can see that the mean scores and the spread of scores vary across teachers. This

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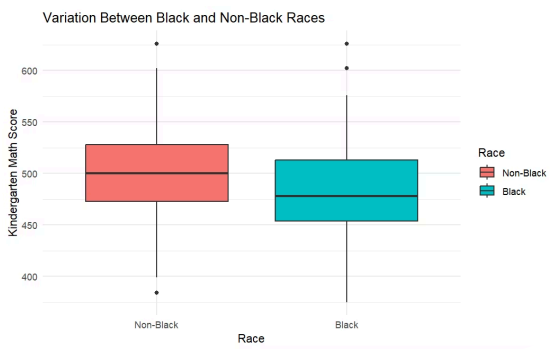
suggests that teacher-level variation may be important. It is also possible that differences in teaching style, classroom environment, or student interactions may influence student performance. So we include Teacher ID in our model to account for this variation.

3. Race and Kindergarten Math Scores



For race, we grouped students into **two** categories because the number of students in the other race groups was relatively small.

Looking at the boxplot above, it suggests that Non-Black students tend to have slightly higher median scores than Black students. The upper quartile for the Non-Black group is also somewhat higher.

This indicates that race is associated with differences in average performance, but it does not completely determine student outcomes.

Race may capture differences in socioeconomic conditions or educational opportunities, so it is reasonable to include race as a **control variable** in the statistical model.

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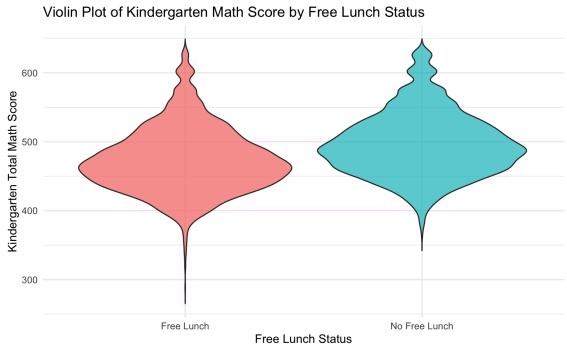
4. Free-Lunch Status and Kindergarten Math Scores



Free-lunch eligibility shows students' socioeconomic background. The figure shows that students who are not eligible for free lunch tend to have higher math scores than those who are eligible. The distribution for the non-free-lunch group is slightly shifted upward, which indicates there are more students with relatively higher scores in this group.

In addition to the difference in average score level, the violin plot also shows the density of the score distributions. Students eligible for free lunch have a wider spread toward lower score values, while students not eligible for free lunch have more density in the middle-to-higher score range. This may suggest that students who are not eligible for free lunch have better family

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economic conditions and may have more learning resources.

We conclude that free-lunch eligibility may reflect socioeconomic conditions that are related to early academic preparation. Therefore, it is reasonable to include free-lunch status as a **control variable** in our statistical model.

Overall, the exploratory analysis suggests that both individual-level factors (**race** and **free-lunch status**) and group-level factors (**school ID** and **teacher ID**) may contribute to variation in students' math performance. Therefore, these variables will be considered in the statistical model.

Proposed Model

Based on our initial exploration of the data, we propose the following statistical model:

$$Y_{ijkls} = \mu + \alpha_i + \beta_j + \gamma_k + \delta_l + u_s + \epsilon_{ijkls}$$

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$$
\begin{aligned}
&Y : \text{Kindergarten total math SAT score} \\
&\mu : \text{Overall mean} \\
&\alpha : \text{Class type (Small, Regular, Regular + Aide)} \\
&\beta : \text{School ID} \\
&\gamma : \text{Race (Black or Other)} \\
&\delta : \text{Free lunch status (Free Lunch, No Free Lunch)} \\
&u : \text{Teacher ID} \\
&\epsilon : \text{Error term}
\end{aligned}
$$

Model Justification

We aim to estimate students' Kindergarten total math SAT score using a linear mixed-effects model. We are interested in interpreting the effect of the various levels of class type on Kindergarten total math SAT scores, so class type is included as a fixed effect. To account for stratified randomized design of the STAR study, school ID is also included as a fixed effect. Since we suspect that the distributions of scores differ meaningfully based on race and free lunch status, we have included these covariates in our model as well. Specifically, we include them as fixed effects; the levels of race and free lunch status in this study are not random. Finally, we include Teacher ID as a random effect; the levels of Teacher ID are random and unknown prior to the experiment. Furthermore, the number of levels of Teacher ID is relatively large, and the effect of Teacher ID on student scores is secondary for the purposes of our analysis.

Model Assumptions

There are several assumptions built into our model.

- Error terms are assumed to be i.i.d. Normal with mean 0 and variance 2

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- Random effects are assumed to be i.i.d Normal with mean 0 and variance u2
- Teacher ID and error terms are mutual independent
- There is no interaction between class type and race and no interaction between class type and free lunch status
- No interference: a student's potential outcome is not affected by other students' assigned class types
- Same version of each treatment: students in all "Small" classes receive the same version of the treatment (same for students in all "Regular" classes and "Regular + Aide" classes)

We decided it was reasonable to assume the absence of interaction between class type and race as well as class type and free lunch status after inspection of the interaction plots (included in the appendix). There was no need to check for interaction between class type and Teacher ID since each teacher only taught one class (and therefore one class type); we are unable to measure how the effect of class type varies within each teacher. Finally, we did not further investigate whether there were interactions between variables excluding class type, since our analysis is focused on the

Inference & Interpretation

Table: Summary of Estimated Coefficients (Fixed Effects)

	Estimated Coefficient	Standard Error	t Value
Intercept	457.289283	11.025474	41.476

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gkclasstypeRegular	-8.438876	2.574114	-3.278
gkclasstypeRegular+Aide	-8.190977	2.559227	-3.201
race_bnbTRUE	-17.056508	2.234509	-7.633
gkfreelunchNotFreeLunch	19.808430	1.267888	15.623

Table: Summary of Estimated Variance (Random Effects)

Groups	Name	Variance	Std. Dev.
gktchid	(Intercept)	266.5	16.33
Residual		1516.3	38.94

Table: Summary of Estimated Differences

Class Size	Estimated Difference	Standard Error	P-Value
Small - Regular	8.439	2.57	0.0030

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Small - Regular + Aid	8.191	2.56	0.0039
Regular - Regular + Aide	-0.248	2.66	0.9952

Estimated Coefficients & Interpretations:

- **Intercept:** The expected Kindergarten math SAT score for a student in a small class who is not Black, attends the school with school ID 112038, and is eligible for free lunch is 457.29.
- **gkclasstypeRegular:** The Kindergarten math SAT score for a student in a regular class is 8.44 points lower than a student in a small class on average, all other variables held constant.
- **gkclasstypeRegular+Aide:** The estimated Kindergarten math SAT score for a student in a regular class is 8.19 points lower than a student in a small class on average, all other variables held constant.
- **race_bnbTRUE:** The estimated Kindergarten math SAT score for a Black student is 17.06 points lower than a student who is not Black on average, all other variables held constant.
- **gkfreelunchNotFreeLunch:** The estimated Kindergarten math SAT score for a student who is not eligible for free lunch is 19.81 points higher than an eligible student on average, all other variables held constant.
- **gktchid:** The ICC ratio is 0.149, which indicates that around 14.9% of variability in Kindergarten math SAT scores can be explained by the random effects from teacher ID.

We observe that on average, students in small classes achieved a score that was 8.44 points higher than students in regu

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and 8.19 points higher than students in regular classes with an aide. Given that the highest Kindergarten total math scale score observed in the data is 626, we will assume that the total score is scored from 0 to 626. This would imply that students in small classes achieved around a 1.35% increase in score compared to students in regular classes and 1.31% increase compared to students in regular classes with an aide. Even if we assumed that the total score is scored starting from 288, the lowest observed score in our dataset, the percent increase would hover around 2% for both regular and regular with aide classes. Therefore, while there is a statistically significant increase in scores among students who were in small classes, the percent increase is relatively small. However, for some educators and parents who prioritize student outcomes, even a marginal increase in these scores is reason enough to advocate for smaller class sizes.

We did not find statistically significant estimated differences in scores between regular classes and regular classes with an aide, which indicates that the effect of having a teaching aide may be minimal for this setting.

Furthermore, based on our model's estimated coefficients, there seems to be a statistically significant difference in scores by race; on average, students who are Black score lower compared to students of other races. There is also a statistically significant positive effect on scores for students who are not eligible for free lunch, enabling us to infer that students coming from low-income families tend to perform worse than their more advantaged peers.

Sensitivity Analysis

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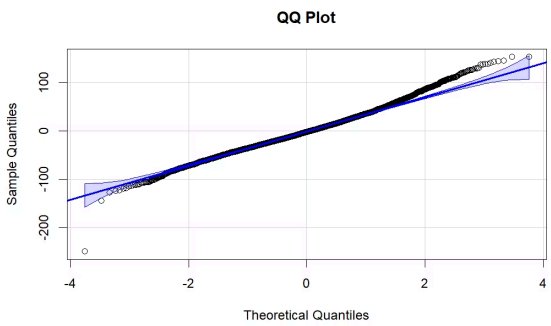
This section analyzes the assumptions for our given ANOVA model, to check how robust our chosen model turned out to be.

1. The QQ Plot of Residuals



The QQ Plot of residuals for our mixed-effects model shows some heaviness in its tail, which suggests some deviance from a normal distribution. However, a majority of the data fits within the theoretical line, with one main section of outliers that breaks from the trend. Given that most of the data satisfies the assumption, and considering that outliers can be common in data observed in the real world, we can mostly choose to assume that normality is met within this model. However, a transformation to the model could still be considered if one wishes to further meet this assumption.

2. Residuals Vs. Fitted Plot

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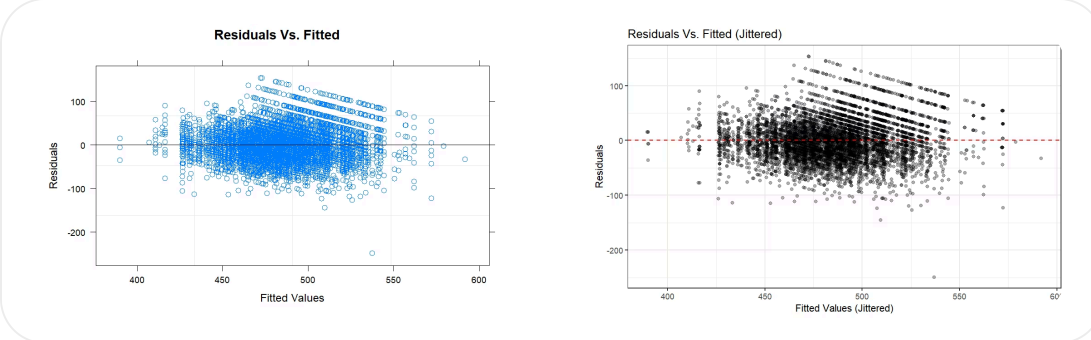

The Residuals Vs Fitted plot shows no obvious concerns in the spread of the data itself, such as a fanning pattern. However, there is a noticeable trend of a few horizontal lines in the top right corner of the plot. To confirm if this pattern was a coincidence, the plot was jittered, which did not show any meaningful change from the previous plot. This suggests that our dependent variable is discrete, and that what is being observed is the "ceiling" of our data (in this case, students getting a perfect score). With this in mind, a different form of the model such as a Tobit model could be considered, that would better account for the SAT scores having a maximum and a minimum that can be earned.

2. Relabeling Class Type Using Class Size

There were some Kindergarten classes that were initially labeled as regular or regular with teacher's aid, but their actual class size dropped into the range for small classes. This would be one example of a potential violation of SUTVA; not all students in "Regular" classes received the same treatment, since some of them actually were in class sizes that were relatively small. To explore how accounting for actual class sizes would affect our conclusions, we re-labeled class types so that classes with 17 students or less were categorized as "Small" classes and then fit another model using the same covariates. The following tables summarize the results.

Table: Summary of Estimated Coefficients (After Relabel

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	Estimated Coefficient	Standard Error	t Value
Intercept	457.51918	11.00507	41.574
gkclasstypeRegular	-9.08674	2.61591	-3.474
gkclasstypeRegular+Aide	-8.21027	2.56373	-3.202
race_bnbTRUE	-17.06417	2.23446	-7.637
gkfreelunchNotFreeLunch	19.80765	1.26785	15.623

Table: Summary of Estimated Differences (After Relabeling)

Class Size	Estimated Difference	Standard Error	P-Value
Small - Regular	9.087	2.62	0.0015
Small - Regular + Aid	8.210	2.56	0.0039
Regular - Regular + Aide	-0.876	2.71	0.9441

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Based on the estimated coefficients and estimated differences, our conclusion does not appear to change. The signs of the estimated coefficients and estimated differences are unchanged, and their magnitudes are relatively unchanged as well. There is still a statistically significant difference in the scores of small classes compared to both regular classes and regular classes with aide, and the difference in scores between regular and regular with aide is still not statistically significant.

👉🏻 **Next: Conclusion**

Conclusion

This look into the STAR project data studied how Kindergarten students' math performance changed based on the size of their classroom. Our main takeaway is that we found an advantage in scores for Kindergarten students in smaller classes compared to those in regular sized classrooms, with and without a teacher aid. While we also compared the differences between regular sized classrooms and regular sized classrooms with teacher aids, we were not able to find any meaningful difference between the two.

It is worth keeping in mind that due to us attempting to maintain the original experimental design to the best of our ability, our findings are only representative of the Kindergarten students in the study, and do not represent the data as a whole. For future studies, analyzing the grades beyond Kindergarten and how they change over time could yield different results from our own. It is also worth noting that the STAR project data was collected starting

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and data was collected in only Tennessee. A more modern dataset that collects data from a wider range of the United States could similarly give us even more insightful results.

However, given the findings we already looked into, we believe that smaller class sizes should be further explored as a means of improving the education system in the United States. Even if it only ends up meaningfully impacting younger students such as Kindergarteners, any advancement is one worth considering. After all, finding potential trends like this is the first step towards improving the educational experience as a whole.

Appendix

Interaction Plots

Made with

R code

Code for basic setup

```
rm(list = ls())
gc()
knitr::opts_chunk$set(echo = TRUE)
knitr::opts_knit$set(root.dir =
"/Users/chichichen/Documents/Master/5-
2026Winter/STA 207/Final_Project")
# ---- 1. Packages ----
pkgs <-
c("haven","dplyr","tidyr","ggplot2","skimr","jani
tor","GGally",

"data.table","plotly","stringr","broom","sandwich
","lmtest",
         "ggalluvial", "forcats", "scales",
"purrr", "tibble")
to_install <- pkgs[!pkgs %in%
installed.packages()[,"Package"]]
```

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```
if(length(to_install)>0)
install.packages(to_install)
invisible(lapply(pkgs, library, character.only =
TRUE))

# ---- 2. Import data ----
# setwd("/Users/chichichen/Documents/Master/5-
2026Winter/STA 207/Final_Project")
df <- read_sav("../STAR_Students.sav") %>%
janitor::clean_names()
cat("Total data count:", nrow(df), "\n\n")

# ---- 3. Variable maps ----
map_long <- list(
  class      = c(k="gkclasstype"),
  school_id  = c(k="gkschid"),
  teacher_id = c(k="gktchid"),
  freelunch  = c(k="gkfreelunch"),
  race   = "race"
)

# ---- 4. Keep only existing columns ----
keep_exist <- function(x, data = df) {
  x[x %in% names(data)]
}

predictor_vars <- map_long %>%
  unlist(use.names = FALSE) %>%
  unique() %>%
  keep_exist(df)

outcome <- "gktmathss"
```

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```
# ---- 5. Build a new dataframe with all selected
variables + outcome ----
analysis_vars <- c(predictor_vars, outcome) %>%
  unique() %>%
  keep_exist(df)

df_analysis <- df %>%
  dplyr::select(all_of(analysis_vars))

# convert factor
df_analysis <- df_analysis %>%
  mutate(across(c("gkclasstype"), as.factor))

# drop rows with missing outcome or missing
teacher ID
df_analysis <- df_analysis %>%
  dplyr::filter(!is.na(gktmathss),
!is.na(gktchid), !is.na(gkfreelunch))

# check result
glimpse(df_analysis)
names(df_analysis)
```

Descriptive Analysis

Distribution of Kindergarten Math Scores by School

```
df_plot <- df %>%
  filter(!is.na(gkschid), !is.na(gktmathss)) %>%
  group_by(gkschid) %>%
  mutate(school_median = median(gktmathss, na.rm
= TRUE)) %>%
```

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```
  ungroup() %>%
  mutate(gkschid =
fct_reorder(as.factor(gkschid), school_median))

ggplot(df_plot, aes(x =gktmathss , y = gkschid ,
fill = school_median)) +
  geom_boxplot(outlier.alpha = 0.4) +
  coord_flip() +
  scale_fill_gradientn(
    colors = c("red", "orange", "yellow",
"green", "blue", "purple")
  ) +
  labs(
    title = "Distribution of Kindergarten Math
Scores by School",
    x = "Kindergarten Total SAT Math Score",
    y ="School ID" ,
    fill = "School median"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle =
90,size=5),
    axis.text.y = element_text(size = 6),
    plot.title = element_text(face = "bold"),
    legend.position = "none"
  )
```

TeacherID vs math score table

```
Teacher_Stats <- df %>%
  group_by(gktchid) %>%
```

Made with

```
  summarise(
    mean = round(mean(gktmathss, na.rm = T), 1),
    median = median(gktmathss, na.rm = T),
    min = min(gktmathss, na.rm = T),
    max = max(gktmathss, na.rm = T),
    Q1 = round(quantile(gktmathss, 0.25, na.rm =
T), 1),
    Q3 = round(quantile(gktmathss, 0.75, na.rm =
T), 1)
  )

Teacher.Stats.Head = head(Teacher_Stats, 10)
Teacher.Stats.Head

Teacher.Stats.Table = kable(Teacher.Stats.Head,
"html") %>%
  kable_styling("striped", full_width = F)

save_kable(Teacher.Stats.Table, file =
"Teacher_Stats_Table.png")
```

Free Lunch Violin plot

```
df_analysis %>%
  filter(!is.na(gkfreelunch), !is.na(gktmathss))
%>%
  mutate(gkfreelunch = factor(gkfreelunch, levels
= c(1, 2), labels = c("Free Lunch", "No Free
Lunch"))) %>%
  ggplot(aes(x = gkfreelunch, y = gktmathss, fill
= gkfreelunch)) +
  geom_boxplot() +
```

Made with

```
  labs(
    title = "Boxplot of Kindergarten Math Score
by Free Lunch Status",
    x = "Free Lunch Status",
    y = "Kindergarten Total Math Score"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Inferential Analysis

Interaction Plots & Model Fitting

```
## Setup
# load libraries
library(tidyverse)
library(haven)
library(lme4)

# read in the data
data = haven::read_sav("STAR_Students.sav")

## Interaction Plots

# create a dataset that only considers
observations that aren't missing any variables of
interest
data_clean = data %>%
  filter(!is.na(gkclasstype)) %>%
  filter(!is.na(gkschid)) %>%
  filter(!is.na(race)) %>%
```

Made with

```
  filter(!is.na(gktchid)) %>%
  filter(!is.na(gkfreelunch)) %>%
  filter(!is.na(gktmathss))

# factor categorical variables
data_clean$gkclasstype =
factor(data_clean$gkclasstype)
data_clean$gkschid = factor(data_clean$gkschid)
data_clean$race = factor(data_clean$race)
data_clean$gktchid = factor(data_clean$gktchid)
data_clean$gkfreelunch =
factor(data_clean$gkfreelunch)

# number of observations
nrow(data_clean)

# create a new variable for race that indicates
Black vs. non-Black
data_clean_2 = data_clean %>%
  mutate(race_bnb = (race==2))

levels(data_clean_2$gkclasstype)

levels(data_clean_2$gkfreelunch)

# rename factor levels
levels(data_clean_2$gkclasstype) = c("Small",
"Regular", "Regular+Aide")
levels(data_clean_2$gkfreelunch) = c("FreeLunch",
"NotFreeLunch")


# interaction plot of race and class type
```

Made with

```
ggplot(data_clean_2, aes(x = gkclasstype, y =
gktmathss, color = race_bnb, group = race_bnb)) +
  stat_summary(fun = mean, geom = "point", size =
3) +
  stat_summary(fun = mean, geom = "line", size =
1) +
  labs(title = "Interaction of Race and Class
Type",
       x = "Class Type",
       y = "Total Math SAT Score",
       color = "Race") +
  scale_color_discrete(labels = c("TRUE" =
"Black", "FALSE" = "Non-Black")) +
  theme_minimal()

# interaction plot of free lunch status and class
type
ggplot(data_clean_2, aes(x = gkclasstype, y =
gktmathss, color = gkfreelunch, group =
gkfreelunch)) +
  stat_summary(fun = mean, geom = "point", size =
3) +
  stat_summary(fun = mean, geom = "line", size =
1) +
  labs(title = "Interaction of Free Lunch Status
and Class Type",
       x = "Class Type",
       y = "Total Math SAT Score",
       color = "Free Lunch Status") +
  theme_minimal()

## Model Training
set.seed(1)
```

Made with

```
# mixed effects model (teacher ID is random
effect)
full_model = lmer(gktmathss ~ gkclasstype +
gkschid + race_bnb + gkfreelunch + (1 | gktchid),
data = data_clean_2)
summary(full_model)

plot(full_model, which = 1, main = "Residuals Vs.
Fitted", xlab = "Fitted Values", ylab =
"Residuals")

library(emmeans)

emmeans(full_model, pairwise~gkclasstype,
adjust='tukey')
```

Model Output

```
Linear mixed model fit by REML ['lmerMod']
Formula:
gktmathss ~ gkclasstype + gkschid + race_bnb +
gkfreelunch +
    (1 | gktchid)
   Data: data_clean_2

REML criterion at convergence: 59330.3

Scaled residuals:
    Min      1Q  Median      3Q     Max
-6.4040 -0.6379 -0.0524  0.5810  3.9456
```

Made with

```
Random effects:
 Groups   Name        Variance Std.Dev.
 gktchid  (Intercept)  266.5   16.33
 Residual             1516.3   38.94
Number of obs: 5852, groups:  gktchid, 325

Fixed effects:
                         Estimate Std. Error t
value
(Intercept)            457.289283  11.025474
41.476
gkclasstypeRegular       -8.438876   2.574114
-3.278
gkclasstypeRegular+Aide  -8.190977   2.559227
-3.201
gkschid123056            12.366694  15.389557
0.804
gkschid128068             9.134396  15.400527
0.593
gkschid128076             0.009713  15.349780
0.001
gkschid128079             1.226406  15.301972
0.080
gkschid130085            14.260628  14.414587
0.989
gkschid159171            51.671883  13.702507
3.771
gkschid161176            10.396734  14.371328
0.723
gkschid161183            46.628100  13.348411
3.493
gkschid162184            26.947798  14.556799
1.851
```

Made with

```
gkschid164198            28.451830  15.394190
1.848
gkschid165199            51.804215  15.459510
3.351
gkschid166203             8.904601  14.447958
0.616
gkschid168211            26.318530  13.755599
1.913
gkschid168214            47.042869  15.389302
3.057
gkschid169219            45.889660  15.581961
2.945
gkschid169229            32.365632  12.978260
2.494
gkschid169231             1.220307  15.361953
0.079
gkschid169280            19.203088  15.396641
1.247
gkschid170295            22.223989  14.516593
1.531
gkschid173312            53.468774  15.344963
3.484
gkschid176329            52.660717  14.507796
3.630
gkschid180344            24.449976  13.740638
1.779
gkschid189378            -0.228447  14.429987
-0.016
gkschid189382             8.228241  14.564327
0.565
gkschid189396             8.612595  15.309981
0.563
gkschid191411            22.481964  15.562205
```

Made with

```
1.445
gkschid193422            35.388825  15.301712
2.313
gkschid193423            40.963528  14.412584
2.842
gkschid201449            42.335429  13.308849
3.181
gkschid203452            12.877714  13.763820
0.936
gkschid203457            32.199844  15.707571
2.050
gkschid205488            12.384234  15.381740
0.805
gkschid205489            11.543030  14.422248
0.800
gkschid205490            -7.307189  15.329550
-0.477
gkschid205491            17.848186  14.472220
1.233
gkschid205492            61.536776  14.366891
4.283
gkschid208501            24.988670  14.546121
1.718
gkschid208503             5.063631  15.321008
0.331
gkschid209510             2.850680  13.795753
0.207
gkschid212522            32.006570  13.793401
2.320
gkschid215533            49.216853  13.285713
3.704
gkschid216536            23.633278  13.730824
1.721
```

Made with

```
gkschid218562           36.679128  14.624651
2.508
gkschid221571          -19.736311  13.714545
-1.439
gkschid221574            2.659519  14.539118
0.183
gkschid225585           12.277023  14.369664
0.854
gkschid228606           43.300347  14.371865
3.013
gkschid230612           57.910826  15.317402
3.781
gkschid231616           47.671664  15.355445
3.105
gkschid234628           27.948545  13.743327
2.034
gkschid244697           12.352264  13.744838
0.899
gkschid244708            7.629287  13.369184
0.571
gkschid244723           13.829009  13.713613
1.008
gkschid244727           33.989729  14.368486
2.366
gkschid244728           11.374748  15.464276
0.736
gkschid244736           70.095845  15.480073
4.528
gkschid244745           56.715374  14.387364
3.942
gkschid244746           46.114740  15.424141
2.990
gkschid244755           54.541278  13.036429
```

Made with

```
4.184
gkschid244764            37.407428  17.327100
2.159
gkschid244774            39.608979  13.740909
2.883
gkschid244776            26.375578  12.852691
2.052
gkschid244780           110.848111  15.330271
7.231
gkschid244796            26.168726  14.659333
1.785
gkschid244799            27.688292  14.538183
1.905
gkschid244801            29.454576  14.381866
2.048
gkschid244806            64.986376  13.034453
4.986
gkschid244818            17.352044  14.340511
1.210
gkschid244831            27.768814  15.276850
1.818
gkschid244839            75.108836  13.821027
5.434
gkschid252885            32.401581  14.457163
2.241
gkschid253888            29.121647  15.612725
1.865
gkschid257899            11.497101  13.715720
0.838
gkschid257905            22.304380  13.055601
1.708
gkschid259915            20.720162  15.383866
1.347
```

Made with

```
gkschid261927           36.974834  14.318081
2.582
gkschid262937           53.345851  14.503628
3.678
gkschid264945           36.523232  13.751989
2.656
race_bnbTRUE           -17.056508   2.234509
-7.633
gkfreelunchNotFreeLunch  19.808430   1.267888
15.623
```

Code for Relabeling Sensitivity Analysis

```
## Sensitivity Analysis: Relabeling

# relabel class types so that regular classes
that dropped into the class size range of small
classes are relabeled as such

data_relabeled_2 = data_clean_2 %>%
  mutate(new_gkclasstype = case_when(
    gkclasssize <= 17 ~ "Small",
    gkclasstype == "Regular" ~ "Regular",
    gkclasstype == "Regular+Aide" ~
"Regular+Aide",
    .default = "0"
  ))

# relevel the factor of relabeled class type
data_relabeled_2$new_gkclasstype =
factor(data_relabeled_2$new_gkclasstype,
levels=c("Small", "Regular", "Regular+Aide"))
```

Made with

```
# train on model on the relabeled data
model_3 = lmer(gktmathss ~ new_gkclasstype +
gkschid + race_bnb + gkfreelunch + (1 | gktchid),
data = data_relabeled_2)
summary(model_3)

emmeans(model_3, pairwise~new_gkclasstype,
adjust='tukey')
```

Output After Relabeling

```
Linear mixed model fit by REML ['lmerMod']
Formula:
gktmathss ~ new_gkclasstype + gkschid + race_bnb
+ gkfreelunch +
    (1 | gktchid)
   Data: data_relabeled_2

REML criterion at convergence: 59329.2

Scaled residuals:
    Min      1Q  Median      3Q     Max
-6.4045 -0.6377 -0.0517  0.5804  3.9447

Random effects:
 Groups   Name        Variance Std.Dev.
 gktchid  (Intercept)  265.1   16.28
 Residual             1516.3   38.94
Number of obs: 5852, groups:  gktchid, 325

Fixed effects:
```

Made with

```
                              Estimate Std. Error
t value
(Intercept)                 457.51918   11.00507
41.574
new_gkclasstypeRegular       -9.08674    2.61591
-3.474
new_gkclasstypeRegular+Aide  -8.21027    2.56373
-3.202
gkschid123056                12.36422   15.35865
0.805
gkschid128068                 9.13077   15.36960
0.594
gkschid128076                 0.01303   15.31874
0.001
gkschid128079                 1.22526   15.27087
0.080
gkschid130085                14.20363   14.38574
0.987
gkschid159171                51.70932   13.67479
3.781
gkschid161176                10.34480   14.34238
0.721
gkschid161183                46.62848   13.32165
3.500
gkschid162184                26.89774   14.52822
1.851
gkschid164198                28.44851   15.36325
1.852
gkschid165199                51.79989   15.42869
3.357
gkschid166203                 8.85419   14.41917
0.614
gkschid168211                26.23394   13.
```

Made with

```
1.911
gkschid168214                47.03991   15.35835
3.063
gkschid169219                45.89976   15.55128
2.952
gkschid169229                32.33260   12.95209
2.496
gkschid169231                 1.22189   15.33099
0.080
gkschid169280                19.20401   15.36573
1.250
gkschid170295                22.17827   14.48791
1.531
gkschid173312                53.46839   15.31394
3.491
gkschid176329                52.61066   14.47907
3.634
gkschid180344                24.36816   13.71307
1.777
gkschid189378                -0.28550   14.40120
-0.020
gkschid189382                 8.18479   14.53567
0.563
gkschid189396                 8.61638   15.27885
0.564
gkschid191411                19.38334   15.55614
1.246
gkschid193422                35.38977   15.27062
2.318
gkschid193423                40.91133   14.38372
2.844
gkschid201449                42.33328   13.28203
3.187
```

Made with

```
gkschid203452                12.91617   13.73623
0.940
gkschid203457                29.04737   15.70063
1.850
gkschid205488                12.39376   15.35077
0.807
gkschid205489                11.48785   14.39341
0.798
gkschid205490                -7.29901   15.29849
-0.477
gkschid205491                17.79473   14.44344
1.232
gkschid205492                61.48607   14.33793
4.288
gkschid208501                24.93636   14.51749
1.718
gkschid208503                 5.06422   15.28993
0.331
gkschid209510                 2.76463   13.76831
0.201
gkschid212522                31.92099   13.76594
2.319
gkschid215533                49.22064   13.25885
3.712
gkschid216536                23.67501   13.70317
1.728
gkschid218562                32.08768   14.66839
2.188
gkschid221571               -19.69234   13.68686
-1.439
gkschid221574                 2.60773   14.51050
0.180
gkschid225585                12.38981   14.34093
```

Made with

```
0.864
gkschid228606                 43.24447   14.34292
3.015
gkschid230612                 57.91352   15.28633
3.789
gkschid231616                 47.66874   15.32444
3.111
gkschid234628                 27.86227   13.71579
2.031
gkschid244697                 12.27438   13.71723
0.895
gkschid244708                  7.63439   13.34248
0.572
gkschid244723                 13.74690   13.68597
1.004
gkschid244727                 33.94034   14.33962
2.367
gkschid244728                  8.33006   15.45900
0.539
gkschid244736                 70.10223   15.44933
4.538
gkschid244745                 56.66714   14.35851
3.947
gkschid244746                 46.11634   15.39331
2.996
gkschid244755                 54.60640   13.01048
4.197
gkschid244764                 37.19430   17.29348
2.151
gkschid244774                 39.52867   13.71333
2.882
gkschid244776                 26.32896   12.82725
2.053
```

Made with

```
gkschid244780                110.85789   15.29923
7.246
gkschid244796                 24.00502   14.65968
1.637
gkschid244799                 27.64082   14.50954
1.905
gkschid244801                 29.40406   14.35295
2.049
gkschid244806                 65.06448   13.00850
5.002
gkschid244818                 17.46868   14.31176
1.221
gkschid244831                 27.76680   15.24572
1.821
gkschid244839                 75.15773   13.79355
5.449
gkschid252885                 32.50179   14.42858
2.253
gkschid253888                 26.09402   15.60644
1.672
gkschid257899                 11.54203   13.68804
0.843
gkschid257905                 22.27387   13.02957
1.709
gkschid259915                 20.72494   15.35290
1.350
gkschid261927                 37.08138   14.28925
2.595
gkschid262937                 53.29374   14.47493
3.682
gkschid264945                 36.43337   13.72448
2.655
race_bnbTRUE                 -17.06417    2.[illegible]
```

Made with

```
-7.637
gkfreelunchNotFreeLunch      19.80765    1.26785
15.623
```

Sensitivity Analysis

Code for Sensitivity Analysis

```
STAR.Model.Residuals = residuals(STAR.Model)
STAR.Model.Fitted = fitted(STAR.Model)

#QQ Plot
STAR.qqPlot = qqPlot(STAR.Model.Residuals, id =
F, main = "QQ Plot", xlab = "Theoretical
Quantiles", ylab = "Sample Quantiles")
```

References & Acknowledgements

References:

https://github.com/Edward-Jing/STA-207

https://web.ma.utexas.edu/users/mks/statmistakes/fixedvsrandom.html

https://web.pdx.edu/~newsomj/mlrclass/ho_randfixd.pdf

https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/SIWH9F

Made with

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Made with